IEOR 221 Assignment 9

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1. We used CRR trees with $N = 2048, 4096, 8192, 16384, 32768$ time steps. For BBSR, we applied Black–Scholes smoothing and Richardson extrapolation, using $2V_{2N} - V_N$ from trees with N and $2N$ time steps. Vega and rho are computed by central finite differences with absolute bumps of 10^{-4}.

CRR

N	NPV	Delta	Gamma	Theta	Vega	Rho
2048	21.1236	0.5218	0.0060	-1.0722	67.1493	85.1271
4096	21.1245	0.5217	0.0060	-1.0718	67.1536	85.1444
8192	21.1249	0.5217	0.0060	-1.0716	67.1550	85.1444
16384	21.1251	0.5217	0.0060	-1.0715	67.1557	85.1448
32768	21.1252	0.5217	0.0060	-1.0715	67.1561	85.1452

BBSR

N	NPV	Delta	Gamma	Theta	Vega	Rho
2048	21.1254	0.5217	0.0060	-1.0715	67.1584	85.1632
4096	21.1254	0.5217	0.0060	-1.0714	67.1564	85.1440
8192	21.1254	0.5217	0.0060	-1.0714	67.1564	85.1450
16384	21.1254	0.5217	0.0060	-1.0714	67.1565	85.1456
32768	21.1254	0.5217	0.0060	-1.0714	67.1565	85.1457

2. For the antithentic method, we used the opposite-sign normal samples as a pair. For the control variate method, we implemented two different control functions, which are the ternimal stock price and the price of corresponding geometric Asian call option.

Method	NPV	MC standard error
Plain MC	5.1914	0.0085
Control variate (terminal stock price)	5.1932	0.0057
Control variate (geometric Asian call)	5.1984	0.0003
Antithetic MC	5.1966	0.0069
Antithetic + control variate (terminal stock price)	5.1981	0.0049
Antithetic + control variate (geometric Asian call)	5.1987	0.0003

Each method uses 2^{20} simulated paths, the fixed seed 0, and is computed on CUDA. For antithetic methods, these paths form 2^{19} pairs. The price of the vanilla call option with the same strike and maturity is 8.8249.